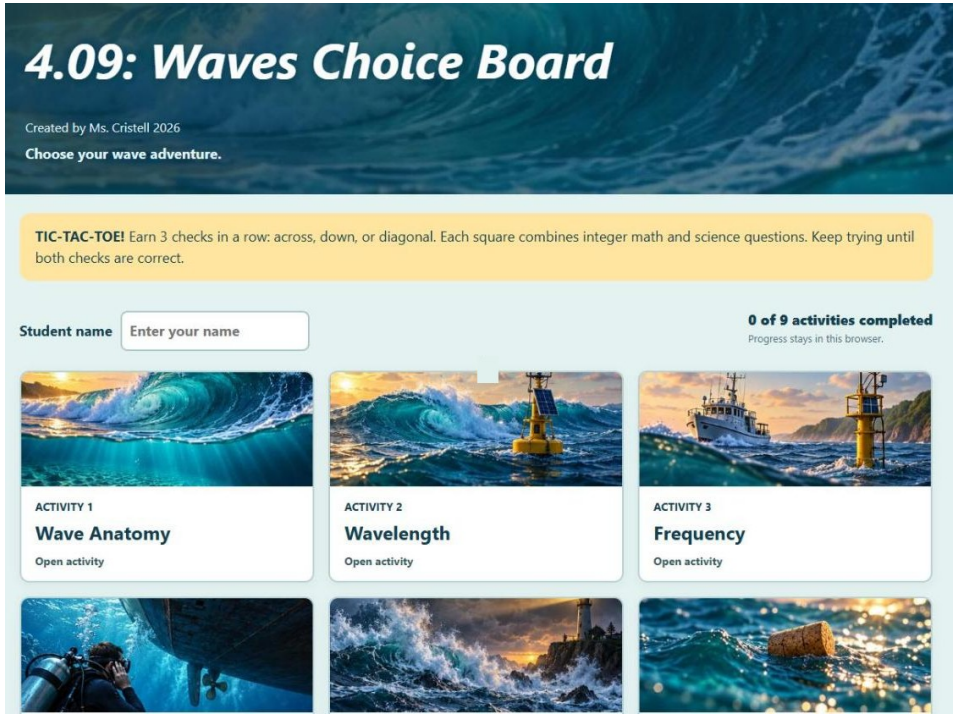
4.09 Review and Catch-Up

Lessons 4.01–4.08: catch up, review, and play



Today's Goals

I can finish any Unit 4 Classkick slides I am missing.

I can explain that waves carry energy, not matter.

I can change amplitude and frequency to match a target wave.

I can use integers to describe waves.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

CASE #2 • TUESDAY

Is the robot less than 25 m above the lake bottom?

WAVE WORD: *Sonar*

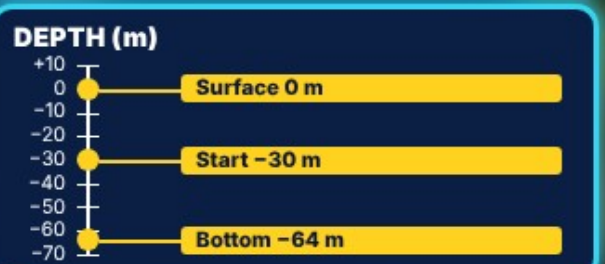
Using sound waves and their echoes to find things underwater.

MYSTERY

An underwater robot uses sonar, a sound wave, to map Lake Erie. It starts 30 meters below the surface, then dives 12 meters deeper. Its sonar finds the lake bottom 64 meters below the surface. Is the robot less than 25 m above the lake bottom?

CLUES

- Start: −30 m
- Dives: −12 m
- Bottom: −64 m





CLAIM

Is the robot less than 25 m above the lake bottom?

☐ YES ☐ NO

EVIDENCE

Solve:

1) −30 + (−12) = _____

2) −42 − (−64) = _____

Rewrite as adding: −42 + _____ = _____

REASONING

Finish the sentence:

Same signs add and keep the sign:

the robot is at _____ m. −42 + _____ = _____.

So the robot is _____ m above the bottom.

Step 1: Catch Up

- [ ] 4.01 Describe a Wave (Parts 1–2)
- [ ] 4.02 Amplitude (Parts 1–2)
- [ ] 4.03 Frequency and Speed + CER + Quest Map
- [ ] 4.04 Refraction + Refraction Lab Map
- [ ] 4.05 Temperature Lab (Parts 1–2) + Graph
- [ ] 4.06 Transverse Waves (Parts 1–2)
- [ ] 4.07 Longitudinal Waves (Parts 1–3) + Quest Map
- [ ] 4.08 EM Spectrum + Escape Room certificate

Not done? Go back and finish it today. **Lower than a B on the 4.03 or 4.07 Quest?** Review your Quest Map, then redo it.

The Big Idea of Unit 4

A wave is a disturbance that carries ENERGY from one place to another. It does NOT carry matter. The medium moves for a moment, then returns to where it started.

Describe a wave by

amplitude, wavelength, frequency, and speed

Mechanical waves

need a medium (sound, water, rope)

EM waves

cross empty space (light, radio, X-rays)

At a boundary, waves can

reflect, refract (bend), or be absorbed

Check yourself: A wave rolls past a buoy on Lake Erie. Does the buoy end up on the beach?

Common Mix-Ups to Avoid

Don't say...	Say...
"The water travels to the shore with the wave."	"The energy travels. The water moves and returns to rest."
"Amplitude is crest to crest."	"Wavelength is crest to crest. Amplitude is rest line to crest."
"A high-frequency wave travels faster."	"In the same medium, all waves travel at the same speed."
"Sound travels fastest through air."	"Sound travels fastest through solids."
"Refraction is when light bounces off a surface."	"That's reflection. Refraction is bending into a new medium."
"Sound can travel through space."	"Sound needs a medium. Only EM waves cross empty space."
"Radio waves are sound waves."	"Radio waves are EM waves. A radio turns them into sound."

Review Poll · 4.01

A student draws a wave on a grid. The crests are at 2 m, 6 m, and 10 m. What is the wavelength?

Single choice poll

A. 4 m

Single choice poll

B. 2 m

Single choice poll

C. 8 m

Single choice poll

D. 10 m

Review Poll · 4.03

At Ohio Stadium, a referee blows a high whistle at the same moment a tuba plays a low note. Both are the same distance from a fan. When does the fan hear them?

Single choice poll

A. Both at the same time

Single choice poll

B. Whistle first: high sounds travel faster

Single choice poll

C. Tuba first: low sounds have more energy

Single choice poll

D. The louder one first

Review Poll · 4.03

A wave travels down a 12 m rope in the gym in 3 seconds. What is its speed?

Single choice poll

A. 36 m/s

Single choice poll

B. 9 m/s

Single choice poll

C. 4 m/s

Single choice poll

D. 15 m/s

Review Poll · 4.04

A laser shines at an angle from air into a glass block. Glass is denser than air. What happens to the light?

Single choice poll

A. Speeds up, bends away from the normal

Single choice poll

B. Same speed, goes straight

Single choice poll

C. Bounces straight back

Single choice poll

D. Slows down, bends toward the normal

Review Poll · 4.06 / 4.08

Two astronauts outside the space station can't hear each other shout, but they can talk by radio. Why?

Single choice poll

A. Radio waves are louder than voices

Single choice poll

B. Sound needs a medium; radio waves are EM waves that cross empty space

Single choice poll

C. Sound is too slow to reach them

Single choice poll

D. Helmets block every wave but radio

Review Poll · CER

Why does a student see lightning before she hears thunder? Claim: light is faster than sound. Evidence: 5 seconds between flash and thunder. Pick the best REASONING.

Single choice poll

A. The flash came 5 seconds before the thunder

Single choice poll

B. Light gets there first because light is faster

Single choice poll

C. Sound must pass energy particle to particle; light is an EM wave and travels much faster

Single choice poll

D. Thunder is louder when lightning is close

Link 1: Wave Combinator

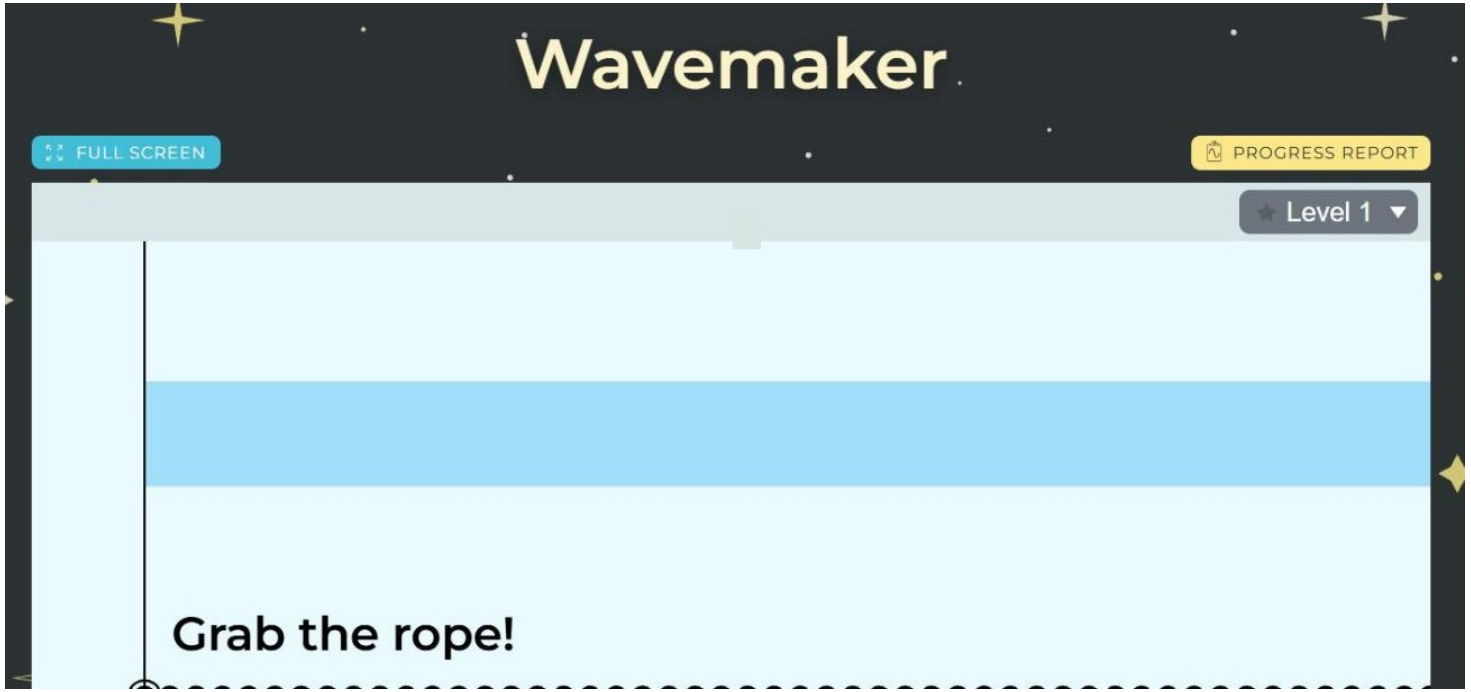


▶️ Open Wave Combinator

Review 4.02 and 4.03

1. Click Play on the Vault.
2. Change the waves to match the target.
3. Fill in the word bank on your Classkick slide.
4. Write the highest level you reached.

Link 2: Wavemaker

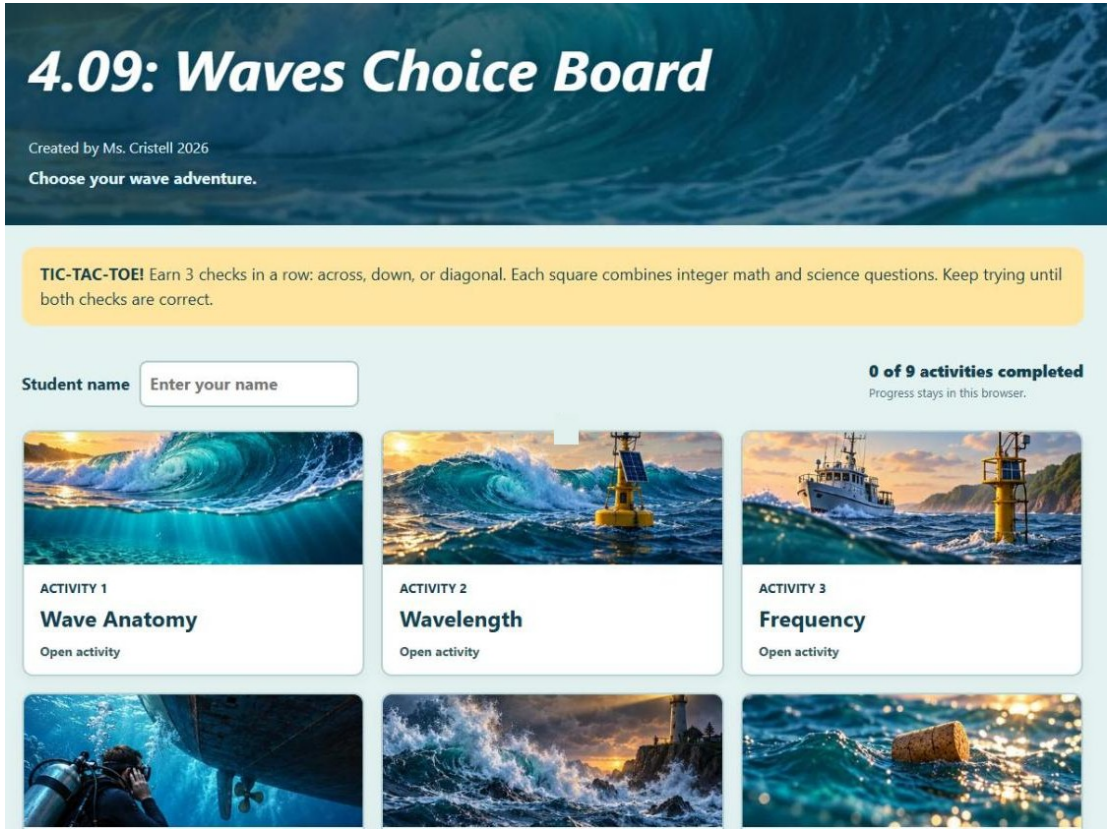


Review 4.02, 4.03, and 4.06

1. Grab the rope.
2. Shake it to match the target shape.
3. Circle what worked on your Classkick slide.

▶️ **Open Wavemaker**

Part 2: Waves Choice Board



▶️ Open the Waves Choice Board

Tic-tac-toe!

1. Type your name.
2. Finish 3 squares in a row: across, down, or diagonal.
3. Each square has 1 integer problem + 1 science question.
4. Snip your Congratulations screen and paste it in Classkick.

Integers and Waves

a) A crest is at +5 cm. A trough is at −5 cm.

Crest − trough = ___ cm

b) A rope moves from +3 cm down to −2 cm.

Change = −2 − (+3) = ___ cm

c) Air warms from −6 °C to +9 °C.

Change = 9 − (−6) = ___ °C

d) Does the warmer air make sound go faster or slower?

Change = end − start. Subtracting a negative = adding. Rest position = 0: above is +, below is −.

Done Early?

Unit 4 Vocab: Blooket

Q7 vocab review

▶️ Open

Unit 4 Vocab: Gimkit

Q7 vocab review

▶️ Open

The Case of the Missing Cell Phone

Decode radio waves by amplitude (4.02, 4.08)

▶️ Open

Bonus Lab: The Particle Wave

Does wavelength change wave speed? (4.03)

▶️ Open



No school Wed–Fri! Use this checklist on Tuesday.



Science Friday! Week 3 · Before break · Tue, Nov 24

My Assignment Checklist

1. Open your gradebook and find each assignment below.
2. Check the box when it is done and graded.
3. **Lower than a B?** Review the Classkick notes for that lesson, then **redo** the assignment.

Done	Assignment (open + assigned this week)	Closes	B or higher?
☐	CLASSKICK The 4 Rules + math Cases #1–#2	With 4.14 Test*	Yes No
☐	CLASSKICK 4.08 slides + Escape Room certificate	With 4.14 Test*	Yes No
☐	CLASSKICK 4.09 Review + Waves Choice Board	With 4.14 Test*	Yes No
☐	CLASSKICK Bonus Lab: The Particle Wave (optional)	With 4.14 Test*	Yes No
☐	GRADED **4.04 Refraction Lab** · 10 pts	Closes Mon, Nov 23 ext: Mon, Nov 30	Yes No
☐	GRADED **4.07 Quest** · 10 pts	Closes Mon, Nov 30 ext: Thu, Dec 3	Yes No

Class Padlet
Videos, slides, websites, and notes for every lesson

All done? Science Fun Padlet
Extra activities for when your work is finished

Coming Up Next Week

Nov 30 – Dec 4 · Red tags show what **closes** that day.

MON 30	4.10 Wave Interference CLOSES: 4.07 QUEST CLOSES: EXT: 4.03 QUEST, 4.04 LAB
TUE 1	4.11 Sound Waves Lab
WED 2	4.12 Explore Sound in a Medium
THU 3	4.13 Unit 4 Review CLOSES: EXT: 4.07 QUEST
FRI 4	Science Friday!



***Classkick reminder!** Your full Unit 4 Classkick packet is **due** with the 4.14 Unit Test (Mon, Dec 14; extended time Mon, Dec 21).

Before You Go

1. Waves carry energy, not matter.
2. Mechanical waves need a medium. EM waves can cross empty space.
3. Frequency up = wavelength down. Same medium = same speed.
4. Finish your catch-up list and the Week 3 checklist.

Re-rate your goals. Happy Thanksgiving! No school Wed–Fri.